

A Unifying Contrast Maximization Framework for Event Cameras, with Applications to Motion, Depth, and Optical Flow Estimation

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Abstract

We present a unifying framework to solve several computer vision problems with event cameras: motion, depth and optical flow estimation. The main idea of our framework is to find the point trajectories on the image plane that are best aligned with the event data by maximizing an objective function: the contrast of an image of warped events. Our method implicitly handles data association between the events, and therefore, does not rely on additional appearance information about the scene. In addition to accurately recovering the motion parameters of the problem, our framework produces motion-corrected edge-like images with high dynamic range that can be used for further scene analysis. The proposed method is not only simple, but more importantly, it is, to the best of our knowledge, the first method that can be successfully applied to such a diverse set of important vision tasks with event cameras.

1. Introduction

Unlike traditional cameras, which produce intensity images at a fixed rate, event cameras, such as the Dynamic Vision Sensor (DVS) [1], have independent pixels that report only intensity changes (called “events”) asynchronously, at the time they occur. Each event consists of the spatio-temporal coordinates of the brightness change (with microsecond resolution) and its sign¹. Event cameras have several advantages over traditional cameras: a latency in the order of microseconds, a very high dynamic range (140 dB compared to 60 dB of traditional cameras), and very low power consumption (10 mW vs 1.5 W of traditional cameras). Moreover, since all pixels capture light independently, such sensors do not suffer from motion blur. In summary, event cameras represent a paradigm shift since visual information is: (i) sampled based on the dynamics of the scene, not based on an external clock², and (ii) encoded us-

¹An animation of the principle of operation of event cameras can be found in the video of [2] <https://youtu.be/LauQ6LWTkxM?t=25>

²If nothing moves in the scene, no events are generated. Conversely, the number of events (samples) increases with the amount of scene motion.

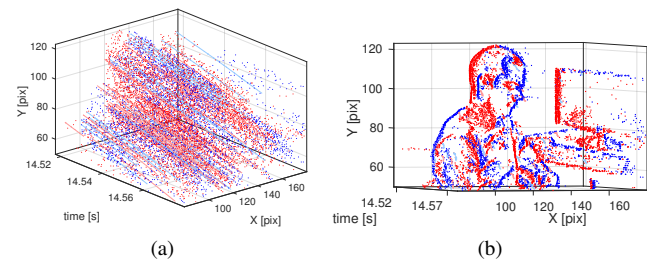


Figure 1: (a) Events (dots) caused by a moving edge pattern and point trajectories in a space-time region of the image plane, colored according to event polarity (blue: positive event, i.e., brightness increase; red: negative event, i.e., brightness decrease). (b) Visualization of the events along the direction of the point trajectories highlighted in (a); corresponding events line up, revealing the edge pattern that caused them. Our approach works by maximizing the contrast of an image of warped events similar to (b). A video demonstrating our framework is available at: <https://youtu.be/KFMZFhi-9Aw>

ing an asynchronous and sparse stream of events, which is fundamentally different from a sequence of images. Such a paradigm shift calls for new methods to process visual information and unlock the capabilities of these novel sensors.

Algorithms for event cameras can be classified according to different criteria. Depending on the way in which events are processed, two broad categories can be distinguished: 1) methods that operate on an *event-by-event* basis, where the state of the system (the estimated unknowns) can change upon the *arrival of a single event* [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15]; and 2) methods that operate on *groups of events*. This category can be further subdivided into two: 2a) methods that *discard temporal information* of the events and accumulate them into frames to re-utilize traditional, image-based computer vision algorithms [16, 17, 18], and 2b) methods that *exploit the temporal information* of the events during the estimation process, and therefore cannot re-utilize traditional computer vision algorithms (sample applications of these methods include variational optical flow estimation [19], event-based multi-

view stereo [20, 21], rotational motion estimation [22], feature tracking [23], pose estimation [24] and visual-inertial odometry [25, 26, 27]).

Event-by-event-based methods rely on the availability of additional appearance data, in the form of grayscale images or a photometric map of the scene, which may be built from past events or provided by additional sensors. Then, each incoming event is compared against such appearance data and the resulting mismatch is used to update the system unknowns. In contrast, methods that operate on groups of events do not rely on prior appearance data. Instead, they aggregate the information contained in the events to estimate the unknowns of the problem. Since each event carries little information and is subject to noise, several events must be processed together to yield a sufficient signal-to-noise ratio for the problem considered.

Both categories present methods with advantages and disadvantages and current research focuses on exploring the possibilities that each category has to offer. Filters, such as the Extended Kalman filter, are the dominant framework of the event-by-event-based type of methods. In contrast, for the groups-of-events-based category, only *ad-hoc solutions* for every problem have been proposed. We present the first unifying framework for processing groups of events while exploiting their temporal information (i.e., category 2b).

Contribution. This paper presents the first unifying framework that allows to tackle several important estimation problems for event cameras in computer vision. In a nutshell, our framework seeks for the point trajectories on the image plane that best fit the event data, and, by doing so, is able to recover the parameters that describe the relative motion between the camera and the scene. The method operates on groups of events, exploiting both their spatio-temporal and polarity information to produce accurate results. In contrast to event-by-event-based approaches, our method does not rely on additional appearance information and it can be used both for estimation problems with very short characteristic time (optical flow) as well as for problems with longer estimation time (monocular depth estimation). Moreover, our framework implicitly handles data association between events, which is a central problem of event-based vision. Additionally, the framework produces motion-corrected event images, which approximate the image gradients that caused the events. These images can serve as input to more complex processing algorithms such as visual-inertial data fusion, object recognition, etc.

The rest of the paper is organized as follows. Section 2 illustrates the main idea behind our framework on a simple example: optical flow estimation. Then, we generalize it and apply it to other problems, such as depth estimation (Section 3.1), rotational motion estimation (Section 3.2) and motion estimation in planar scenes (Section 3.3). Section 4 concludes the paper.

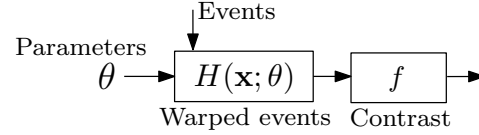


Figure 2: Events are warped according to point trajectories described by motion parameters θ , resulting in an image of warped events $H(\mathbf{x}; \theta)$. The contrast of H measures how well events agree with the candidate point trajectories.

2. Contrast Maximization Framework

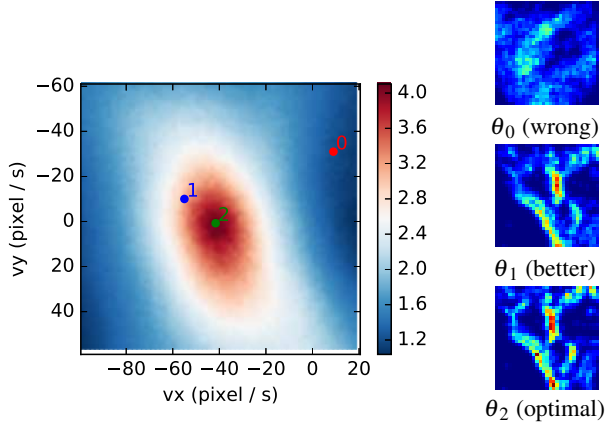
Since event cameras report only pixel-wise brightness changes, it implies that, assuming constant illumination, there must exist (i) relative *motion* between the camera and the objects in the scene and (ii) sufficient texture (i.e., brightness gradients) for events to be generated. Hence, event cameras respond to the apparent motion of edges. In the absence of additional information about the appearance of the scene that caused the events, the problem of extracting information from the events becomes that of establishing correspondences between them, also known as data association, i.e., establishing which events were triggered by the same scene edge. Since moving edges describe point trajectories on the image plane, we expect corresponding events to be triggered along these trajectories. Fig. 1 illustrates this idea with a simple example where the point trajectories are nearly straight lines. We propose to find the point trajectories that best fit the event data, as in Fig. 1b. Let us describe our framework using a simple yet important example (optical flow estimation) and then let us generalize it to other estimation problems.

2.1. Example: Optical Flow Estimation

Assume that we are given a set of events $\mathcal{E} \doteq \{e_k\}_{k=1}^{N_e}$ in a spatio-temporal neighborhood of a pixel, as in Fig. 1a, and the goal is to estimate the optical flow (i.e., motion vector) at that pixel based on the information contained in the events. Recall that each event $e_k \doteq (x_k, y_k, t_k, p_k)$ consists of the space-time coordinates of a predefined brightness change together with its polarity $p_k \in \{-1, +1\}$ (i.e., the sign of the brightness change).

As is standard, optical flow is measured over a small time interval (ideally infinitesimal), and the trajectories followed by points on the image plane are locally straight, approximated by translations: $\mathbf{x}(t) = \mathbf{x}(0) + \mathbf{v}t$, where $\mathbf{x} \doteq (x, y)^\top$ and $\mathbf{v}$ is the velocity of the point (i.e., the optical flow). Hence, we expect corresponding events (triggered by the same edge) to lie on such trajectories (Fig. 1b).

Our framework, summarized in Fig. 2, consists of counting the events or summing their polarities along the straight trajectories given by a candidate optic flow and computing the variance (f) of the resulting sums (H), which measures how well the events agree with the candidate trajectories.



(a) $f(\theta)$ as a function of the optical flow $\theta \equiv \mathbf{v}$. For illustration, we show in (b) the associated images of warped events H for three candidate velocities. (b) Warped events $H(\mathbf{x}; \theta)$, for θ_i , $i = 0, 1, 2$ in (a).

Figure 3: *Optical Flow* (patch-based) estimation. The correct optical flow can be found as the one that maximizes the contrast (Fig. 3a) of the images of warped events (Fig. 3b).

More specifically, to sum the events, we warp them to a reference time t_{ref} using the proposed trajectories. Following the local-flow-constancy hypothesis [28], we assume that the flow is constant in the space-time neighborhood spanned by the events, and warp the events, $e_k \mapsto e'_k$, according to

$$\mathbf{x}'_k \doteq \mathbf{W}(\mathbf{x}_k, t_k; \theta) = \mathbf{x}_k - (t_k - t_{\text{ref}})\theta, \quad (1)$$

with $\theta = \mathbf{v}$ the candidate velocity. Then, we build an image patch of warped events:

$$H(\mathbf{x}; \theta) \doteq \sum_{k=1}^{N_e} b_k \delta(\mathbf{x} - \mathbf{x}'_k), \quad (2)$$

where each pixel $\mathbf{x}$ sums the values b_k of the warped events $\mathbf{x}'_k$ that fall within it (indicated by the Dirac delta δ). If $b_k = p_k$, the polarities of the events along the trajectories are summed; whereas, if $b_k = 1$, the number of events along the trajectories are computed. In the rest of the paper, we use $b_k = 1$ and show comparisons with $b_k = p_k$ in the supplementary material. Finally, we compute the variance of H , which is a function of θ ,

$$f(\theta) = \sigma^2(H(\mathbf{x}; \theta)) \doteq \frac{1}{N_p} \sum_{i,j} (h_{ij} - \mu_H)^2, \quad (3)$$

where N_p is the number of pixels of $H = (h_{ij})$ and $\mu_H \doteq \frac{1}{N_p} \sum_{i,j} h_{ij}$ is the mean of H .

Fig. 3a shows the variance (3) as a heat map, for a given set of events $\mathcal{E}$. Observe that it is smooth and has a clear peak. The images of warped events corresponding to three different motion vectors θ_i in Fig. 3a are displayed in Fig. 3b (represented in pseudo-color, from blue

(few events) to red (many events)). As can be seen, the warped events are best aligned in the image H that exhibits highest variance, i.e., highest contrast (and sharpness), attained at $\theta^* = \arg \max_{\theta} f(\theta) \approx (-40, 0)^\top$ pixel/s. Hence, our strategy for optical flow estimation consists of seeking the parameter θ that maximizes (3).

As anticipated in Section 1, our method produces motion-corrected image patches H , which approximate the gradients of the brightness pattern that caused the events (also illustrated in Fig. 1b). More specifically, H represents the brightness increment along the candidate trajectories. For optimal trajectories, this increment is proportional to $\nabla I \cdot \mathbf{v}$, the dot product of the brightness gradient and the motion vector (due to the optical flow constraint). These motion-corrected image patches can be useful for feature tracking, such as [27, 29].

Our framework implicitly defines data association between the events. Replacing the delta in (2) with a smooth approximation, $\delta(\mathbf{x}) \approx \delta_\varepsilon(\mathbf{x})$, such as a Gaussian, $\delta_\varepsilon(\mathbf{x} - \mu) \doteq \mathcal{N}(\mathbf{x}; \mu, \varepsilon^2 \text{Id})$ (we use $\varepsilon = 1$ pixel), we see that every warped event e'_k influences every other event e'_n , and the amount of influence is given by $\delta_\varepsilon(\mathbf{x}'_n - \mathbf{x}'_k)$, which, in the case of a Gaussian, is related to the Euclidean distance $\|\mathbf{x}'_n - \mathbf{x}'_k\|$. Hence, our method has a built-in soft data association between all events, implicitly given by a function of the distance between them: the further away warped events are, the less likely they are corresponding events.

2.2. General Description of the Framework

The example in the previous section contains all the ingredients of our event-processing framework. Let us now describe it in a more generic manner, to apply to other estimation problems (in Section 3).

We propose to find the point trajectories on the image plane that best fit the event data. More specifically, assume we are given a set of events, $\mathcal{E} \doteq \{e_k\}_{k=1}^{N_e}$, typically continuous in time, as in Fig. 1, acquired while the camera and/or the scene undergo some motion for which we have a geometric model of how points move on the image plane. Such a geometric model depends on the particular estimation problem addressed (optical flow, depth estimation, motion estimation). The goal is to estimate the parameters of the model based on the information contained in the events. We assume that estimation is possible, in that the model parameters (unknowns) are shared among multiple events (fewer parameters than events) and are observable. To solve the problem, we build candidate point trajectories $\mathbf{x}(t)$ according to the motion and scene models, and measure the goodness of fit between these trajectories and the event data $\mathcal{E}$ using an objective function (3) (see Fig. 2). Then, we use an optimization algorithm to seek for the point trajectories (i.e., the parameters θ of the motion and scene models) that maximize the objective function. As shown in Fig. 3b, good

trajectories are those that align corresponding events, and so the objective function that we propose measures how well events are aligned along the candidate trajectories. There are two by-products of our framework: (i) the estimated point trajectories implicitly establish correspondences between events (i.e., data association), (ii) the trajectories can be used to correct for the motion of the edges.

2.2.1. Steps of the Method. Our method consists of three main steps:

1. Warp the events into an image H , according to the point trajectories defined by the above-mentioned geometric model and candidate parameters θ .
2. Compute a score f based on the image of warped events.
3. Optimize the score or objective function with respect to the parameters of the model.

In step 1, events are geometrically transformed taking into account their space-time coordinates and other known quantities of the point-trajectory model, $e_k \mapsto e'_k(e_k; \theta)$, resulting in a set of warped events $\mathcal{E}' \doteq \{e'_k\}_{k=1}^{N_e}$. The warp, such as $\mathbf{W}$ in (1), transports each event along the point trajectory that passes through it, until a reference time is reached (e.g., the time of the first event): $e_k \doteq (x_k, y_k, t_k, p_k) \mapsto (x'_k, y'_k, t_{\text{ref}}, p_k) \doteq e'_k$.

In step 2, an image or histogram of warped events $H(\mathcal{E}')$ is created (using their polarities p_k or their count), and an objective function (a measure of dispersion) is computed, $f(H(\mathcal{E}'))$. We use as dispersion metric the variance of H , which is known as *contrast* in image processing terminology, and we seek to maximize it. The objective function represents the statistics of the warped events $\mathcal{E}'$ according to the candidate model parameters θ , hence it measures the goodness of fit of θ to the event data $\mathcal{E}$.

In step 3, an optimization algorithm, such as gradient ascent or Newton’s method, is applied to obtain the best model parameters, i.e., the point trajectories on the image plane that best explain the event data. The framework is flexible, not relying on any specific optimizer.

2.2.2. Contrast Maximization. By maximizing the variance of the image of warped events $H(\mathcal{E}'(\theta))$ we favor the point trajectories that accumulate (i.e., align) the warped events on the image plane. The accumulation of warped events in some regions and the dispersion of events in other regions (since the total number of events N is constant) produces an image H with a larger range, and, therefore a higher contrast, which is clearly noticeable if H is displayed in grayscale, and hence the name *contrast maximization* framework. In essence, the goal of the optimization framework is to “pull apart” the statistics (e.g., polarity) of the regions with and without events, in a similar way to the segmentation approach in [30].

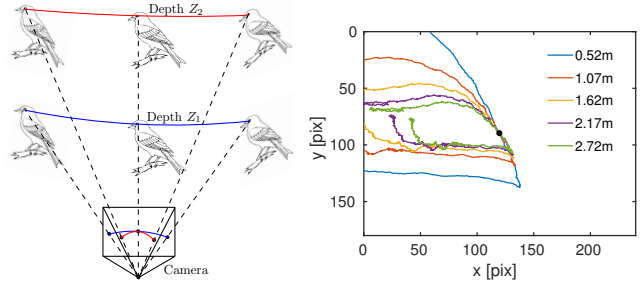


Figure 4: *Depth Estimation.* Left: trajectories of a 3D point (the eye of the bird) relative to the camera. The point closer to the camera has larger apparent motion, and therefore typically describes longer trajectories on the image plane. Right: Trajectories of an image point (the image center, in black), for different depth values with respect to a reference view while the camera undergoes a 6-DOF motion. Each depth value produces a different point trajectory; yielding a 1D family of curves. This is an example of a 2 s segment from a sequence of the Event Camera Dataset [31].

Contrast is related to image sharpness, and therefore, an observed effect is that the images of warped events with higher contrast are also sharper (see Fig. 3b), which is consistent with the better alignment of the warped events. The image of warped events associated to the optimal parameters is a motion-corrected edge-like image, where the “blur” (trace of events) due to the moving edges has been removed (cf. Fig. 3b top and bottom). Such edge-like image represents the brightness-increment patterns causing the events.

2.2.3. Computational Complexity. The core of our method is the computation of the image of warped events (2), whose computational complexity is linear on the number of events to be warped. The computation of the contrast (3) is typically negligible compared to the effort required by the warp. The overall cost of the method also depends on the choice of algorithm used to maximize the contrast, which is application-dependent.

3. Sample Applications

Our framework is flexible and generic. In this section, we use it to solve various important problems in vision.

3.1. Depth Estimation

Consider the scenario of an event camera moving in a static scene and the goal is to infer depth. That is, consider the problem of event-based multi-view stereo (EMVS) [21, 32] (3D reconstruction) from a set of events $\mathcal{E} = \{e_k\}_{k=1}^{N_e}$. By assumption of the problem, the pose of the event camera $P(t)$ is known for every time t , where P denotes the projection matrix of the camera. We assume that the intrinsic

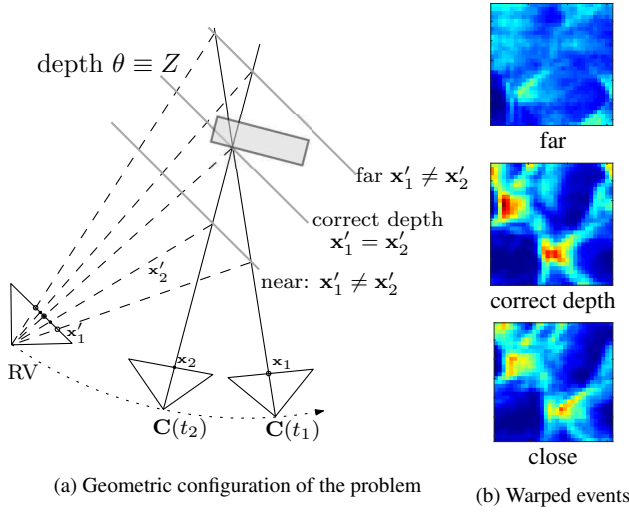


Figure 5: *Depth Estimation*. Alignment of warped events $\mathbf{x}'_k(\theta)$, for different depth values $\theta \equiv Z$, measured with respect to the reference view (RV). In (b), the patch of warped events (2) is represented in pseudo-color, from few events aligned (blue) to many events (red). At the correct depth, the patch has the highest variance (i.e., image contrast).

parameters of the camera are also known³ and that lens distortion has been removed. Following the framework in the previous section, we first specify the geometry of the problem, the warp (i.e., point trajectories) and the score function.

In this scenario, the trajectory of an image point obtained by projection of a 3D point is parametrized by the known 6-DOF motion of the camera and the depth of the 3D point with respect to a reference view. This yields a 1D family of curves in the image plane, parametrized by depth. Each depth value gives a different curve followed by the point in the image plane. This is illustrated in Fig. 4. As in the Space-Sweep approach of [21], we consider a reference view provided by a virtual camera at some location, for example, at a point along the trajectory of the event camera (see Fig. 5). Let us formulate the problem for a small patch in the reference view. For simplicity, we assume that all the points within the patch have the same depth, which is some candidate value $\theta = Z$. The three main steps of our method (Section 2.2.1) are as follow:

1. (a) Transfer the events (triggered at the image plane of the moving event camera) onto the reference view using the candidate depth parameter, as illustrated in Fig. 5. An event e_k is transferred, using the warp ($\mathbf{W}$), onto the

event $e'_k = (\mathbf{x}'_k, t_{\text{ref}}, p_k)$ with

$$\mathbf{x}'_k = \mathbf{W}(\mathbf{x}_k, \mathbf{P}(t_k), \mathbf{P}_v; \theta), \quad (4)$$

where $\mathbf{P}(t)$ is the pose of the event camera at time t and $\mathbf{P}_v \doteq \mathbf{P}(t_{\text{ref}})$ is the pose of the virtual camera. The warp is the same as in space-sweep multi-view stereo: points are transferred using the planar homography [35, ch. 13] induced by a plane parallel to the image plane of the reference view and at the given depth (see Fig. 5).

(b) Create an image (patch) of warped events (2) by counting the number of events along the candidate point trajectories (e.g., Fig. 4).

2. Measure the goodness of fit between the events and the depth value θ by means of the variance (i.e., contrast) of the image of warped events (3).
3. Maximize the contrast by varying the depth parameter θ .

Fig. 5 illustrates the above steps. In Fig 5a an event camera with optical center $\mathbf{C}(t)$ moves in front of a scene with an object (gray box). Two events $e_i = (\mathbf{x}_i, t_i, p_i)$, $i = \{1, 2\}$ are transferred from the event camera to a reference view (RV) via the warp (4) using three candidate depth values (in front of, at the object and behind it, respectively). The points transferred using depth values in front and behind the object are not aligned, $\mathbf{x}_1 \neq \mathbf{x}_2$, whereas the points transferred using the correct value are aligned, $\mathbf{x}_1 = \mathbf{x}_2$. Event alignment is more noticeable in Fig. 5b, with patches of warped events (2) from a sequence of the Event Camera Dataset [31]. At the correct depth, the warped events present a better alignment, and therefore higher contrast (and sharpness) of H (here represented in pseudo-color instead of grayscale), compared to the cases of wrong depth values (labeled as “close” and “far”).

Fig. 6 shows depth estimation for two patches in a sequence from the dataset [31]. The sequence was recorded with a DAVIS camera [33], which outputs asynchronous events and grayscale frames (at 24 Hz). The frame in Fig. 6a is only shown for visualization purposes; our method processes solely the events. Fig. 6b shows how the contrast of the warped events (3) varies with respect to the depth parameter θ , for each patch in Fig. 6a. The actual warped events (2) at selected depth values are displayed in Fig. 6c. Remarkably, the contrast curves (Fig. 6b) have a smooth variation, with a clear maximum at the correct depth value. The warped events in (2) show, indeed, a better alignment (i.e., higher contrast) at the correct depth than at other depth values. As in Fig. 5b, a pseudo-color scale is used to represent the pixels of H , from few events aligned (blue) to many events (red). Additionally, note that, although the contrast curves are from different patches, the “spread” of the curves increases with the value of the peak depth, which is consistent with the well-known fact that, in short-baseline stereo, depth uncertainty grows with depth. Finally, observe that

³An event camera may be intrinsically calibrated using event-based algorithms, such as [2] or, as in the case of the DAVIS [33], using standard algorithms [34] since the DAVIS comprises both a traditional frame camera and an event sensor (DVS) in the same pixel array.

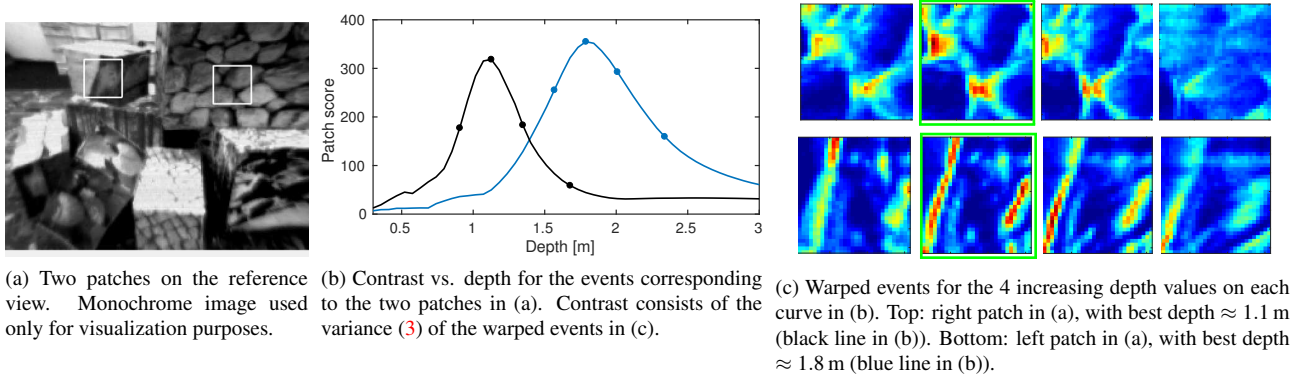


Figure 6: *Depth estimation* by contrast maximization. Two patches are analyzed.

the patches of warped events with highest contrast (highlighted in green in Fig. 6c), resemble the edgemap (gradient magnitude) of the grayscale patches in Fig. 6a, showing that indeed the events are triggered by moving edges and that our framework recovers, as a by-product, an approximation to edgemap of the scene patch that caused the events.

The above procedure yields the depth value at the center of the patch in the reference view. Repeating the procedure for every pixel in the reference camera for which there is sufficient evidence of the presence of an edge produces a semi-dense depth map. This is shown in Fig. 7, using a sequence from the dataset [31]. In this experiment, we computed the contrast (3) on patches of 3×3 pixels in the reference view; the patches H were previously weighted using a Gaussian kernel, $H(\mathbf{x}) \leftarrow w(\mathbf{x})H(\mathbf{x})$, to emphasize the contribution of the center pixel. A map of the maximum contrast for every pixel of the reference view is shown in Fig. 7b. For visualization purposes, contrast is represented in negative form, from bright (low contrast) to dark (high contrast). This map is used to select the points in the reference view with largest contrast, which are the points for which depth estimation is most reliable. The result is a semi-dense depth map, which is displayed in Fig. 7a, color coded, overlaid on a grayscale frame. To select the points, we used adaptive thresholding [36, p.780] on the contrast map, and we used a median filter to remove spike noise from the depth map. As it can be observed, depth is most reliable at strong brightness edges of the scene. The depth map is visualized as a color-coded point cloud in Figs. 7c and 7d.

If the images of warped events (2) are interpreted as depth slices of the disparity space image (DSI) formed by event back-projection [21], our framework estimates depth by selecting DSI regions with largest local variance (contrast). Moreover, our framework allows to iteratively refine the depth values (e.g., by gradient ascent on the objective function (3)) so that they are continuous, i.e., they are not constrained to be the discrete set of values imposed by a voxelization of the DSI.

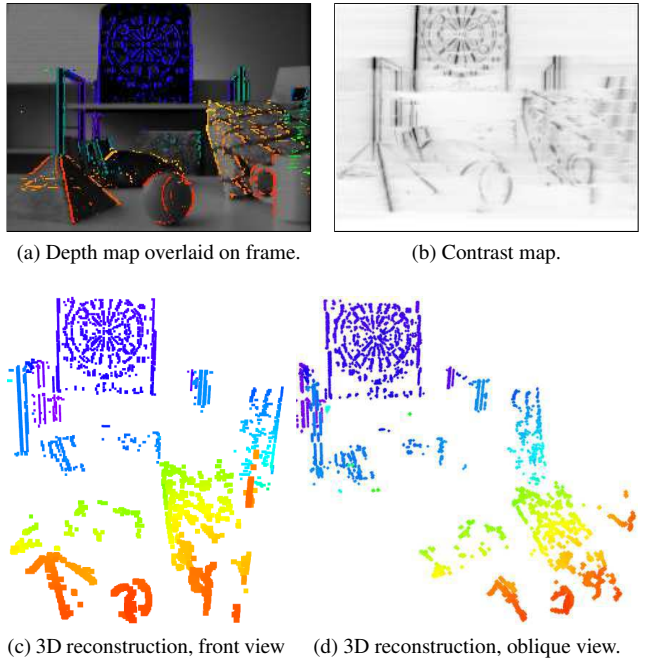
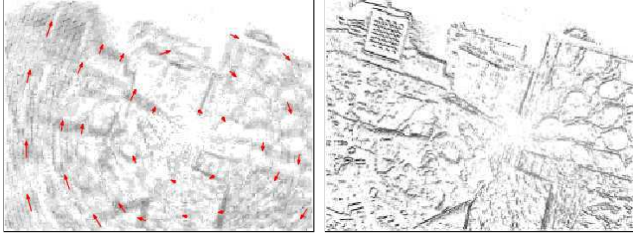


Figure 7: *Depth Estimation*. 3D reconstruction on the *slider_depth* sequence of the dataset [31]. One million events in a timespan of 3.4 s were processed. In (a), (c) and (d), depth from the reference view is color-coded, from red (close) to blue (far), in the range of 0.45 m to 2.4 m. In the supplementary material we show how the reconstruction changes with the number of events processed.

3.2. Rotational Motion Estimation

Our framework can also be applied to the problem of rotational motion estimation [9, 22, 37, 38]. Consider the scenario of an event camera rotating in a static scene and the goal is to estimate the camera’s ego-motion using the events. As in Section 3.1, assume that the camera is calibrated (known intrinsic parameters and no lens distortion).



(a) Warped events using $\theta = \mathbf{0}$. Red arrows indicate the true motion of the edges causing the events. (b) Warped events using estimated angular velocity θ^* , which produces motion-corrected, sharp edges.

Figure 8: *Rotational Motion Estimation*. Images of warped events, displayed in grayscale to better visualize the motion blur due to event misalignment and the sharpness due to event alignment. The direction of the rotation axis is clearly identifiable on the left-image as the point with least motion blur and where fewer events are triggered. Dataset [31].

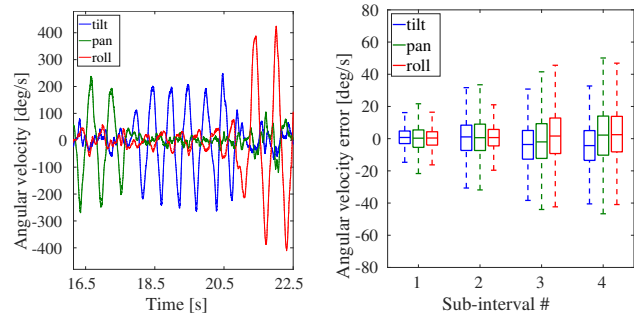
The type of motion and the motion parameters themselves enforce constraints on the trajectories that image points can follow. For example, in a rotational motion with constant velocity, all point trajectories are parametrized by 3-DOFs: the angular velocity. Our framework aligns events by maximizing contrast over the set of admissible trajectories: those compatible with the rotational motion.

Let us specify the steps of the method (Section 2.2.1) for the problem at hand. Consider all events $\mathcal{E}$ over a small temporal window $[0, \Delta t]$; small enough so that the angular velocity ω can be considered constant within it, and let $t_{\text{ref}} = 0$. In calibrated coordinates, image points transform according to $\bar{\mathbf{x}}(t) \propto \mathbf{R}(t)\bar{\mathbf{x}}(0)$, where $\bar{\mathbf{x}} \propto (\mathbf{x}^\top, 1)^\top$ are homogeneous coordinates and $\mathbf{R}(t) = \exp(\hat{\omega}t)$ is the rotation matrix of the (3D) motion [35, p. 204]: $\exp$ is the exponential map of the rotation group $SO(3)$ [39] and $\hat{\omega}$ is the cross-product matrix associated to ω . In step 1, events are warped to t_{ref} according to the point-trajectory model: $\mathbf{x}'_k = \mathbf{W}(\mathbf{x}_k, t_k; \theta)$, with $\theta = \omega$ the angular velocity and

$$\mathbf{W}(\mathbf{x}_k, t_k; \theta) \propto \mathbf{R}^{-1}(t_k)\bar{\mathbf{x}}_k = \exp(-\hat{\theta}t_k)\bar{\mathbf{x}}_k. \quad (5)$$

The image of warped events is then given by (2). Approaches like [22] use the event polarity, which is indeed beneficial if the motion is monotonic (i.e., does not change direction abruptly). However, as we show, polarity is not needed (see Figs. 8 and 9). In steps 2 and 3, the objective function (3) is maximized using standard optimization algorithms such as non-linear conjugate gradient [40].

Figure 8 shows the result of our method on a group $\mathcal{E}$ of $N_e = 30000$ events acquired while the camera is rotating approximately around its optical axis. As it can be seen, our method estimates the motion parameters that remove the motion blur from the image of warped events, providing the sharpest image. Fig. 9 reports the accuracy of our method



(a) Angular velocity: estimated (solid) vs. ground truth (dashed). (b) Angular velocity error: estimated vs. ground truth for 4 subintervals of 15 s.

Figure 9: *Rotational Motion Estimation*. Accuracy evaluation. The `boxes_rotation` sequence of the dataset [31] contains 180 million events and reaches speeds of up to $670^\circ/\text{s}$. Comparatively, our method produces small errors.

using the same dataset and error metrics as [22]. Both plots compare the recovered rotational motion of the event camera against ground truth. Fig. 9a shows the estimated and ground-truth motion curves, which are almost indistinguishable relative to the magnitude of the motion. Fig. 9b further analyzes the error between them in four subintervals of 15 s with increasing angular velocities (and therefore increasing errors). Our method is remarkably accurate, with RMS errors of approximately $20^\circ/\text{s}$ with respect to peak excursions of $670^\circ/\text{s}$, which translates into 3 % error. Moreover, our approach does not need a (panoramic) map of the scene to estimate the rotational motion, as opposed to approaches [9, 38]. It also does not need to estimate optical flow prior to fitting a 3D motion, as in [41]. In a nutshell, our approach acts like a visual event-based gyroscope.

3.3. Motion Estimation in Planar Scenes

In this section, we show how our framework can be applied to the problem of motion estimation under the assumption of a planar scene (i.e., planar homography estimation), which allows to extract the ego-motion parameters of the camera (rotation and translation) as well as the parameters of the plane containing the scene structure.

In this scenario, image points transform according to $\bar{\mathbf{x}}(t) \propto \mathbf{H}(t)\bar{\mathbf{x}}(0)$, where $\bar{\mathbf{x}} \propto (\mathbf{x}^\top, 1)^\top$ are homogeneous coordinates and $\mathbf{H}(t)$ is a 3×3 homography matrix. For simplicity, we use $t = 0$ as reference time, and so $\mathbf{H}(0) = \text{Id}$ is the identity. The point trajectories described by $\mathbf{x}(t)$ have the same number of DOFs as $\mathbf{H}(t)$, which, for short time intervals in which we consider $\mathbf{H}$ to be constant, is 8-DOF. We aggregate events along the point trajectories $\mathbf{x}(t)$ defined by candidate homographies $\mathbf{H}(t)$, and maximize the contrast of the resulting image of warped events to recover the homography that best explains the event data.

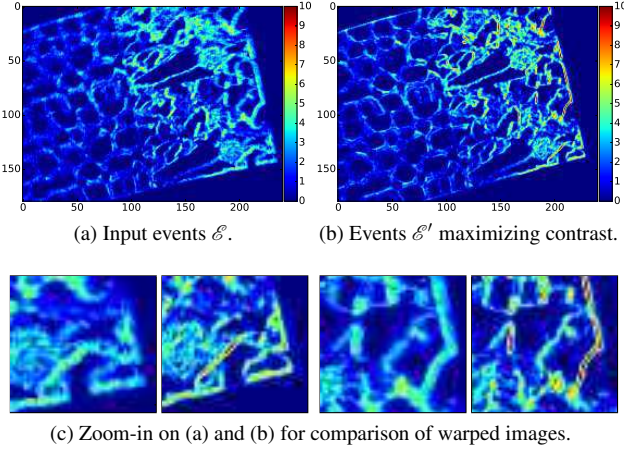


Figure 10: *Motion Estimation in Planar Scenes*. (a) Input events, with each pixel counting the number of events triggered; (b) Warped events using the planar homography parameters $\theta = \{\omega, \mathbf{v}/d, \mathbf{n}\}$ that maximize image contrast: $\omega = (0.086, 0.679, 0.439)^\top$, $\mathbf{v}/d = (0.613, -0.1, 0.333)^\top$, $\mathbf{n} = (0.07, 0.075, -0.995)^\top$.

In case of a homography induced by a plane with homogeneous coordinates $\pi = (\mathbf{n}^\top, d)^\top$, we have [35] $H(t) \propto R(t) - \frac{1}{d}\mathbf{t}(t)\mathbf{n}^\top$, where $P(0) \propto (\text{Id}|\mathbf{0})$ is the projection matrix of the reference view and $P(t) \propto (R(t)|\mathbf{t}(t))$ is the projection matrix of the event camera at time t . For $t \in [0, \Delta t]$ in a short time interval, we may compute $P(t)$ from the linear and angular velocities of the event camera, $\mathbf{v}$ and ω , respectively, by assuming that they are constant within the interval: $R(t) = \exp(\hat{\omega}t)$ and $\mathbf{t}(t) = \mathbf{v}t$. Hence, we may parametrize $H(t) \equiv H(t; \theta)$ by $\theta = (\omega^\top, \mathbf{v}^\top/d, \phi, \psi)^\top \in \mathbb{R}^8$, where the 2-DOFs (ϕ, ψ) parametrize the unit vector of the plane $\mathbf{n}$ (e.g., latitude-longitude parameters). The parameters $\mathbf{v}/d$ account for the well-known fact that, without additional information, there is a scale ambiguity: the decomposition of a planar homography [35, 39] only provides the direction of the translation, but not its magnitude.

Hence, we consider the events $\mathcal{E}$ in a short time interval, $[0, \Delta t]$, and map them onto the image plane of the reference view using the warp specified in calibrated coordinates by

$$\mathbf{W}(\mathbf{x}_k, t_k; \theta) \propto H^{-1}(t_k; \theta) \bar{\mathbf{x}}_k. \quad (6)$$

Then, the image of warped events is built as usual (2), and its variance (3), i.e., contrast, is computed to assess the quality of the parameters θ on event alignment.

Fig. 10 shows our method in action. The scene consists of a freely moving event camera viewing a rock poster [31]. In Fig. 10a, a set $\mathcal{E}$ of $N_e = 50000$ events is displayed in an event image, with each pixel counting the number of events triggered within it, i.e., as if the identity warp ($\mathbf{x}'_k = \mathbf{x}_k$) was used in (2). Fig. 10b shows the result of contrast

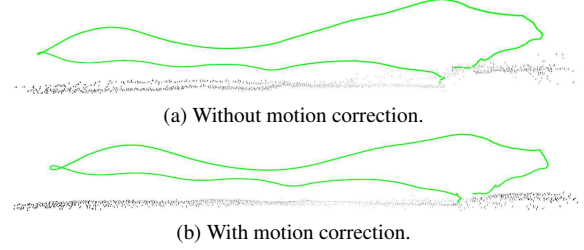


Figure 11: *Motion Estimation in Planar Scenes*. Scene structure (black dots) and camera motion (green trajectory) obtained by a visual-inertial algorithm [27], with and without motion-corrected event images.

maximization: for the optimal parameters θ , the warped events are better aligned with each other, resulting in an image (2) with higher contrast than that in Fig. 10a. Observe that event alignment by contrast maximization produces a motion-corrected image, which is specially noticeable at texture edges: in Fig. 10a (no motion correction) edges are blurred, whereas in Fig. 10b edges are sharp.

Fig. 11 shows another example of our framework. In this sequence, an event camera is hand-held, looking downwards while a person is walking outdoors over a brick-patterned ground. Event images are used in a visual-inertial algorithm [27] that recovers the trajectory of the event camera and a sparse 3D point map of the scene. The motion-corrected images resulting from homography estimation (cf. Fig. 10b) produce better results, which can be seen by the more flat point cloud representing the floor in the scene.

4. Conclusion

In this work, we have focused on showing the capabilities of our framework to tackle several important vision problems for event cameras (3D reconstruction, optical flow and motion estimation), which we believe is its most remarkable characteristic. We showed that there is a simple, principled way to process events in all these problems in the absence of additional appearance information about the scene: maximizing event alignment along point trajectories on the image plane.

Event cameras have multiple desirable properties: low latency, very high dynamic range and low power consumption. We believe this work is a significant step forward into leveraging the advantages of these novel sensors in real-world scenarios, overcoming the limitations of conventional imaging sensors.

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